Supernova-type shell wrinkling applied to the CoE setup

Expanding $H_0 = 70$ run: point mass M at the CoE, shells start with $v = k\,r$, $k = 160$ km s^{-1}Mpc^{-1}, run time $T = 7.86$ Gyr, observer at $D = 3.2$ Gpc, view to 787 Mpc. $1\times = 6.08 \times 10^{51}$ kg (original geometry: $D = 58.2$ Gpc, shell $R = 14.3$ Gpc, $1\times = 7.5 \times 10^{52}$ kg; same dimensionless run, so all lengths below scale by $58.2/3.2 = 18.2$).

1. Shell motion and effective gravity

$$\frac{d^2r}{dt^2} = -\frac{GM}{r^2}, \qquad r(0) = r_0, \quad \dot{r}(0) = k\,r_0, \qquad r(T) = D$$

$$g_{\rm eff}(t) = -\frac{d^2r}{dt^2} = \frac{GM}{r(t)^2}$$

$r(t)$: radius of our shell from the CoE; r_0: its start radius; G: gravitational constant; $g_{\rm eff}$: deceleration felt in the shell frame (points outward, so the dense shell sits "on top").

2. Rayleigh–Taylor growth

$$\sigma(t) = \sqrt{A\,k_\lambda(t)\,g_{\rm eff}(t)}, \qquad A = \frac{\rho_{\rm shell} - \rho_{\rm in}}{\rho_{\rm shell} + \rho_{\rm in}} \approx 1, \qquad k_\lambda(t) = \frac{2\pi}{\lambda}\,\frac{D}{r(t)}$$

$$N(\lambda) = \int_0^T \sigma(t)\,dt \qquad \text{(e-folds; amplitude grows by } e^N\text{)}$$

σ: growth rate; A: Atwood number; $\rho_{\rm shell}, \rho_{\rm in}$: shell and interior densities; λ: wavelength measured today; k_λ: physical wavenumber of a ripple that keeps a fixed angle on the stretching shell.

3. Vishniac thin-shell mode

$$\lambda_{\rm V} \approx (2\text{–}5)\,\Delta, \qquad \ell = \frac{2\pi R}{\lambda_{\rm V}}, \qquad \text{needs } \frac{\rho_{\rm shell}}{\rho_0}10,\ \gamma \to 1$$

Δ: shell thickness; R: shell radius; ℓ: angular mode number; ρ_0: density ahead of the shell; γ: adiabatic index. No shell thickness is defined in our earlier work, so $\Delta/R = 0.01$–0.2 is scanned (Bertschinger 1983 cosmological blast shells: $\Delta \approx 0.03\,r_s$).

4. Key numbers

	1×	4×	10×
start radius r_0 (Gpc)	1.45	1.57	1.76
$g_{\rm eff}$ start / now (10^{-11} m s^{-2})	20 / 4.2	69 / 17	138 / 42
shell speed now (km/s)	209,000	171,000	110,000
e-folds N at 40 / 100 / 150 Mpc	6.3 / 4.0 / 3.3	12 / 7.4 / 6.0	17 / 10.5 / 8.5
e-folds at 1000 Mpc	1.3	2.3	3.3
wavelength with $N = 3$ (Mpc)	180	610	1200

Vishniac spacing ($\lambda \approx 3\Delta$) on the shell at $D = 3.2$ Gpc: $\Delta/R = 0.01 \to 96$ Mpc ($\ell \approx 210$), $0.03 \to 290$ Mpc ($\ell \approx 70$), $0.1 \to 960$ Mpc, $0.2 \to 1900$ Mpc. BAO (150 Mpc) needs $\Delta/R \approx 0.016$; walls (100 Mpc) need ≈ 0.01; voids (40 Mpc) need ≈ 0.004. On the $R = 14.3$ Gpc shell every $\Delta/R \geq 0.01$ gives 430 Mpc.

RT has no built-in scale: growth rises toward short wavelengths, so a cutoff (pressure, viscosity, shell thickness) must pick the spacing. Gravity at 1× gives only a few e-folds at 100–150 Mpc; 10× gives about ten.





